# Supplementary Information for: MBD-ML: Many-body dispersion from machine learning for molecules and materials

## S1. MBD-ML TRAINING AND PARAMETERS

The MBD-ML model is based on the SO3LR/SO3krates architecture [1, 2]. For the purpose of the MBD-ML model, the architecture was extended to predict $C_6^r$ and $\alpha_0^r$ ratios. The training was conducted on the QCML dataset [3] for which a combined loss function including these ratios with equal weighting was employed, using a robust loss function with $\alpha = 1$. The learning rate was adapted exponentially using the AMSGrad optimizer with a learning rate of 0.001 and a decay rate of 0.9 applied every 1M steps. Compared to the original SO3LR model, MBD-ML was kept relatively light weight, with a feature dimension of 64, 2 layers, 4 heads, a maximal degree of 4 and 32 radial basis functions. Thus, with a short-range cutoff radius of 4.0 Å and 2 message-passing layers, MBD-ML exhibits an effective receptive field of 8.0 Å.

## S2. SELECTION OF DATA SUBSETS

**DES370k [4]** We used the dimer structures of the DES370k set, which were recomputed at the PBE0+MBD-NL level of theory as part of the recently published QCell biomolecular dataset [5].

**OMol25 [6]** For the construction of the OMol25 subset used in this study, we chose the structures of the SPICE and the PDB protein pocket fragments subset, as the union of those two contains molecules ranging from 3 to 350 atoms. To achieve a uniform selection of system sizes, the molecules of these two sets were binned by the number of atoms, using bins of size 10 ($1-10$ atoms, $11-20$ etc.) and a total of 883 molecules was randomly picked in a uniform fashion.

**OMC25 [7]** To ensure a maximally diverse subset for the present work, each of the 200 molecular crystals that were picked feature a unique molecule i.e., there are no two polymorphs of the same molecular crystal. Other than that the structures were chosen randomly without additional constraints.

**OMat24 [8]** Starting from the OMat24 validation set, 6 materials categories were defined based on the band gap and the number of atoms in the unit cell:

1. `small-metal`
2. `small-insulator`
3. `intermediate-metal`
4. `intermediate-insulator`
5. `large-metal`
6. `large-insulator`,

where `metal` and `insulator` structures were defined by the PBE band gap of the original OMat24 entry to be less than 0.5 eV and $\geq 0.5$ eV, respectively, while `small`, `intermediate` and `large` structures were made up of at most 50 atoms, at most 100 atoms and over 100 atoms, respectively. For each of these 6 categories, up to 50 materials were randomly selected and recomputed at the HSE06+MBD-NL level of theory, of which 203 calculations were completed successfully and used to benchmark the MBD-ML model in this work.

## S3. *AB INITIO* CALCULATIONS

All DFT re-calculations of the OMol25, DES370k, OMC25 and OMat24 datasets were performed using the FHI-aims electronic structure package [9, 10] (version 231212.1). All molecules and molecular crystal structures, including the subsets of the OMol25, DES370k and OMC25 datasets were computed at the PBE0+MBD-NL level of theory, while the inorganic material structures from the OMat24 dataset were computed at the HSE06+MBD-NL level. For the convergence of the SCF-cycle the default accuracy settings in combination with default tight basis sets were used. Relativistic effects were included using the scalar atomic ZORA method. For both hybrid functionals an MBD damping parameter $\beta$ of 0.83 was used. For all molecular crystals, k-meshes with a spacing of 0.03 Å$^{-1}$ were employed after finding that for a random subset of 50 OMC25 molecular crystal structures this yielded forces converged to below 1 meV/Å. For the inorganic materials, the k-mesh was chosen based on the PBE band gap provided in the original OMat24 dataset. For materials with a PBE band gap of less than 0.5 eV, a very fine k-mesh spacing of 0.0125 Å$^{-1}$ was employed, while materials with larger band gaps were recomputed at the HSE06+MBD-NL level of theory with a coarser k-spacing of 0.02 Å$^{-1}$.

## S4. EXCLUSION OF ANIONS FROM QCML

### S4.1. Basis set convergence for anions

For the QCML subset of negatively charged molecules, we identified outliers in the ratios $\alpha_0^r$ and $C_6^r$ in the original data. These outliers correspond to atoms with unusually large values, reaching 20–60, whereas both ratios typically lie within the 0–2 range. In most cases, the anomalous values occur for hydrogen atoms, which is particularly unexpected.

A detailed analysis of the underlying *ab initio* calculations revealed that, for anions, $\alpha_0^r$ and $C_6^r$ are highly sensitive to the cut_pot parameter in FHI-aims. This parameter defines the onset of a steeply increasing confining potential $v_{\mathrm{cut}}(r)$ used to localize numerical atom-centered radial functions. For more details on $v_{\mathrm{cut}}(r)$ and its role in constructing basis functions, we refer the reader to Section 3 of the original FHI-aims paper [9].

In practical terms, cut_pot, together with the width parameter w (kept fixed here), determines the spatial support of a basis function. The default tight settings for organic elements (cut_pot = 4 Å, w = 2 Å) yield a total radial extent of 6 Å, which is generally sufficient to obtain meV-accurate total energies. However, the accurate description of diffuse electron density in anions requires explicit convergence tests with respect to cut_pot.

We randomly selected several anions exhibiting anomalous ratios and performed calculations varying cut_pot between 3 and 8 Å, while keeping all other computational parameters consistent with QCML settings [3]. The total energies of the anions show strong sensitivity to cut_pot, varying by up to ±1.5 eV relative to the default 4 Å setting (Fig. S1a). Convergence is slow, indicating significant contributions from slowly decaying density tails. By contrast, neutral molecules of comparable size (benzene, pentane) exhibit variations of only about 2 meV between cut_pot = 4 Å and 8 Å.

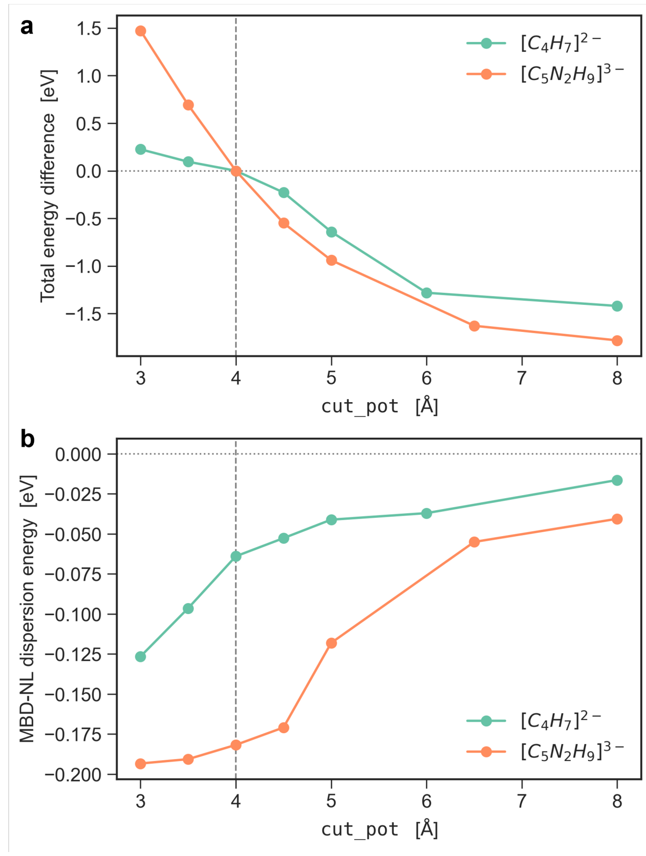


**Fig. S1.** Convergence of (**a**) total energies and (**b**) MBD-NL dispersion energies with respect to the cut_pot parameter for two example anions. Total energies are plotted as the differences relative to the default 4 Å setting (marked by a vertical dashed line).

The corresponding MBD-NL dispersion energies are likewise unconverged and increase unphysically as the basis becomes more extended (Fig. S1b). This behavior is further illustrated by plotting $\alpha_0^{\mathrm{r}}$ and $C_6^{\mathrm{r}}$ for hydrogen atoms versus cut_pot, which reveals a dramatic increase in both ratios with increasing basis extent (Fig. S2). Hydrogen atoms are emphasized because they are typically located at the molecular periphery and thus most sensitive to density tails, although similar trends are observed for some heavier atoms.

Physically, this divergence arises because a more extended basis permits slower decay of the electron density, and the VV polarizability functional is highly sensitive to density gradients in the tail region [11]. The performance of VV and MBD-NL for anions was not assessed in the original work, and this issue has not previously been reported.

Further analysis shows that the problematic species are actually thermodynamically unstable, as indicated by their computed electron affinities (Table S1). For an $N$-electron system, the electron affinity is defined as $\mathrm{EA}(N) = E(N) - E(N+1)$, with positive values indicating stability of the $(N+1)$-electron state. We evaluated EAs by sequentially removing electrons from the charge states present in QCML. Calculations employed the long-range corrected LC-$\omega$PBEh functional ($\omega = 0.2$ Bohr$^{-1}$) [12] as implemented in Q-Chem [13], together with aug-cc-pVQZ basis sets to ensure adequate diffuseness [14].

As summarized in Table S1, only singly charged anions are stable in all four cases; higher charge states are clearly unstable. Consequently, such pathological entries must be recomputed with physically meaningful charge states to form a reliable subset of negatively charged molecules within QCML. To mitigate similar issues in future datasets, additional screening could be applied, for example using HOMO energies as a proxy for stability. As shown in Table S1, HOMO values obtained with a long-range corrected functional correlate well with EAs, consistent with the ionization potential theorem applied to the $(N+1)$-electron system, which yields $\mathrm{EA}(N) = \mathrm{IP}(N+1) = -\mathrm{HOMO}(N+1)$ for the exact functional [15].

A more comprehensive investigation, including systematic benchmarking of basis sets and exchange–correlation functionals, is warranted but lies beyond the scope of the present work. Until such validation and filtering are performed, we do not recommend using QCML data for negatively charged molecules, particularly dispersion energies and derived ratios.

**TABLE S1.** Total energies, electron affinities and HOMO levels (all in eV) for various charged states of the four considered molecules. The first row in each case corresponds to the charged state included in QCML.

| $C_4H_7$ | | | | $C_2NH_4$ | | | |
|---|---|---|---|---|---|---|---|
| $Q$ | $E_{\text{tot}}$ | EA | HOMO | $Q$ | $E_{\text{tot}}$ | EA | HOMO |
| -2 | -4252.28 | -2.81 | 2.79 | -2 | -3621.08 | -3.36 | 3.29 |
| **-1** | **-4255.09** | **0.96** | **-1.00** | **-1** | **-3624.44** | **1.61** | **-1.58** |
| 0 | -4254.13 | | -9.15 | 0 | -3622.84 | | -9.07 |

| $C_5N_2H_9$ | | | | $C_6O_2H_9$ | | | |
|---|---|---|---|---|---|---|---|
| $Q$ | $E_{\text{tot}}$ | EA | HOMO | $Q$ | $E_{\text{tot}}$ | EA | HOMO |
| -3 | -8290.99 | -4.16 | 4.76 | -3 | -10441.5 | -4.74 | 4.75 |
| -2 | -8295.15 | -2.86 | 2.42 | -2 | -10446.2 | -2.62 | 2.71 |
| **-1** | -8298.02 | **1.67** | **-1.72** | **-1** | **-10448.9** | **1.30** | **-1.82** |
| 0 | -8296.35 | | -7.04 | 0 | -10447.6 | | -8.77 |

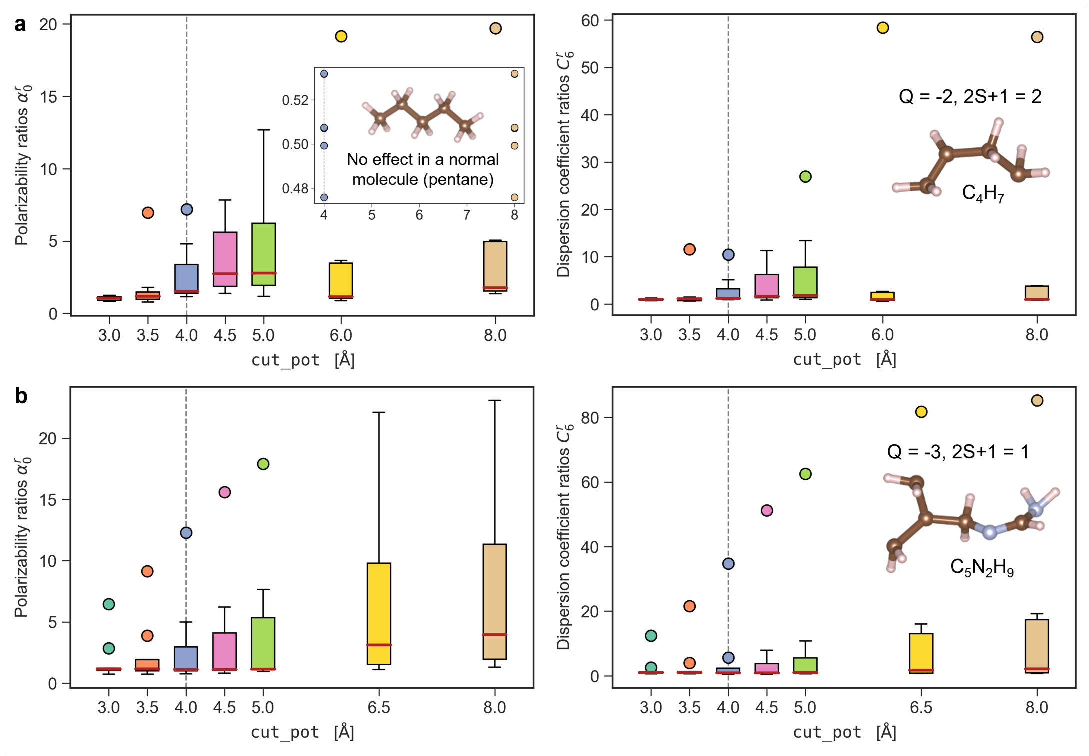


**Fig. S2.** Boxplots of $\alpha_0^r$ (left) and $C_6^r$ (right) with varying `cut_pot` for hydrogen atoms in the two negatively charged molecules: (**a**) $[C_4H_7]^{2-}$ and (**b**) $[C_5N_2H_9]^{3-}$ (the structures shown in the insets). The inset in the upper left panel illustrates the $\alpha_0^r$ ratios in pentane molecule for `cut_pot` $= 4$ Å and 8 Å.

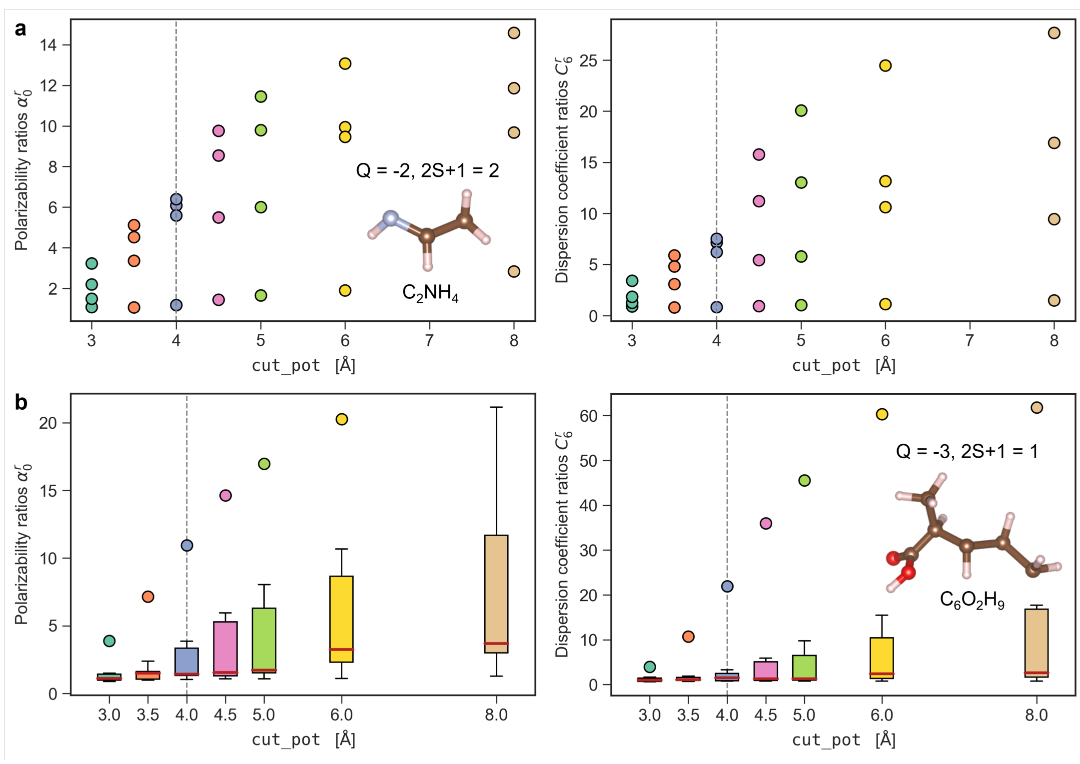


**Fig. S3.** The same as Fig. S2 but for (**a**) $[C_2NH_4]^{2-}$ and (**b**) $[C_6O_2H_9]^{3-}$ (the structures shown in the insets).

### S4.2. Performance of MBD-ML with anions included

If these pathological anions from the QCML dataset are included in the accuracy measurement of the MBD-ML model, one finds numerous outliers in the MBD ratio prediction (see Fig. S4), which is plausible due to the insufficient computational settings for negatively charged molecules outlined in the previous subsection.

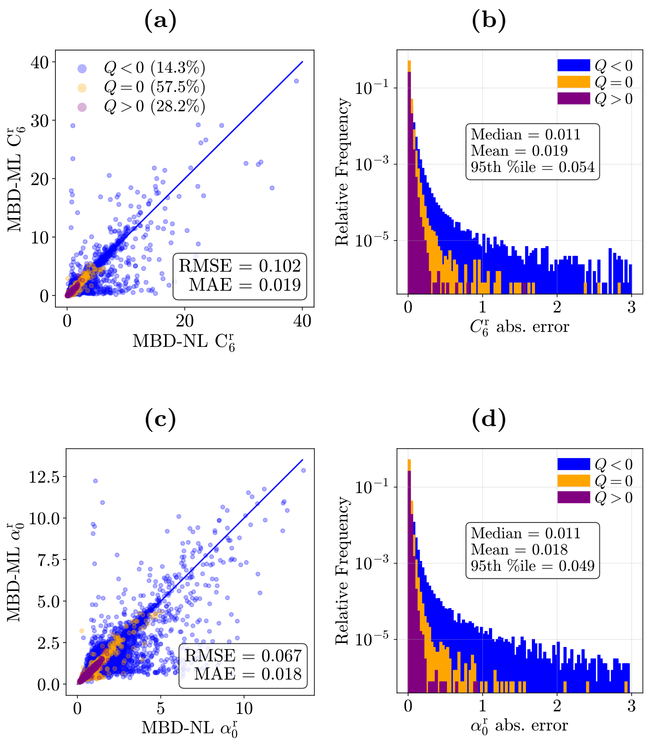


**Fig. S4.** Performance of PBE0+MBD-ML in predicting the $C_6$ and $\alpha_0$ ratios of the QCML test set including all negatively charged molecules.

We find, however, that the resulting MBD energy and force contributions are basically unaffected from these outliers (see Fig. S5).

## S5. MBD-ML VS MBD-NL PERFORMANCE FOR OMOL25 AND DES370K SUBSETS

While the main analysis of the MBD-ML performance on the OMol25 and DES370k subsets excluded alkali and alkaline earth elements, here we additionally show the accuracy of the model if those elements are included relative to the *ab initio* MBD-NL reference. Figs. S6 and S7 illustrate the performance of the MBD-ML model on the OMol25 subset with and without alkali and alkaline earth elements. Figs. S8 and S9 do so analogously for the

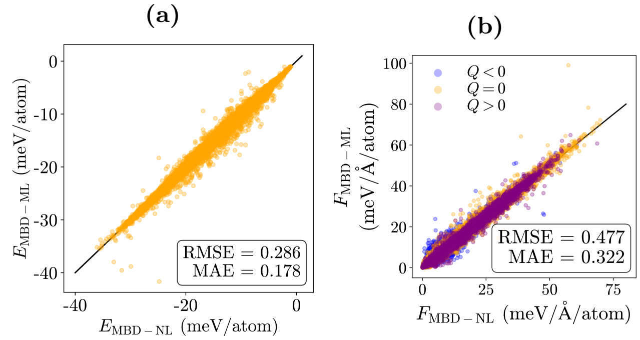


**Fig. S5.** Performance of PBE0+MBD-ML in predicting the MBD contribution to the total energy and atomic forces of the QCML test set including all negatively charged molecules.

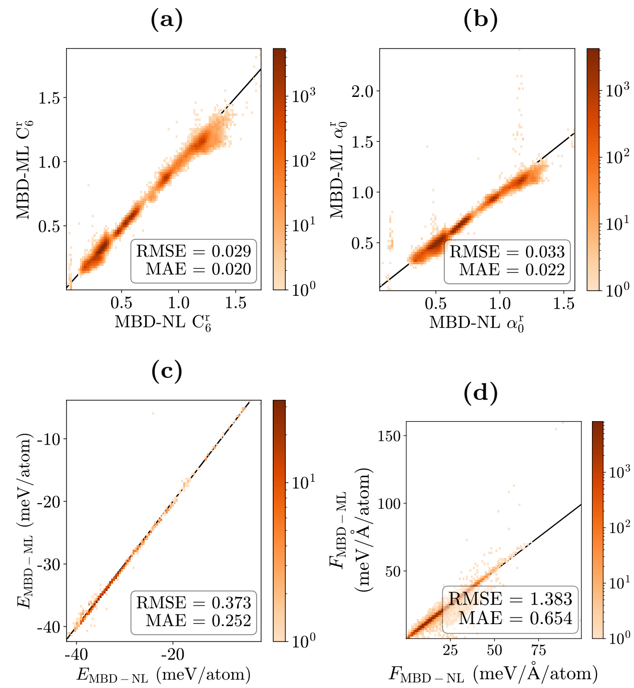


**Fig. S6.** Performance of PBE0+MBD-ML in predicting the $C_6$ and $\alpha_0$ ratios and the MBD contribution to the total energy and atomic forces of the OMol25 test set including alkali and alkaline earth elements.

DES370k test set.

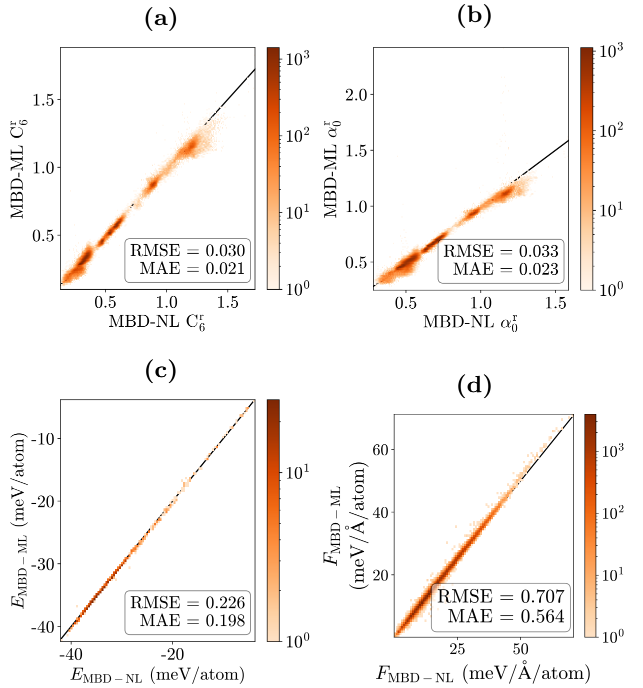


**Fig. S7.** Performance of PBE0+MBD-ML in predicting the $C_6$ and $\alpha_0$ ratios and the MBD contribution to the total energy and atomic forces of the OMol25 test set excluding alkali and alkaline earth elements.

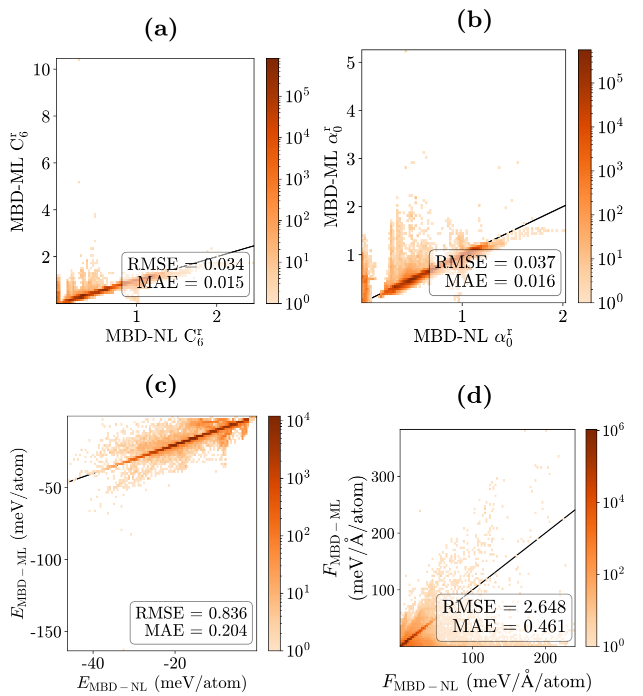


**Fig. S8.** Performance of PBE0+MBD-ML in predicting the $C_6$ and $\alpha_0$ ratios and the MBD contribution to the total energy and atomic forces of the DES370k test set including alkali and alkaline earth elements.

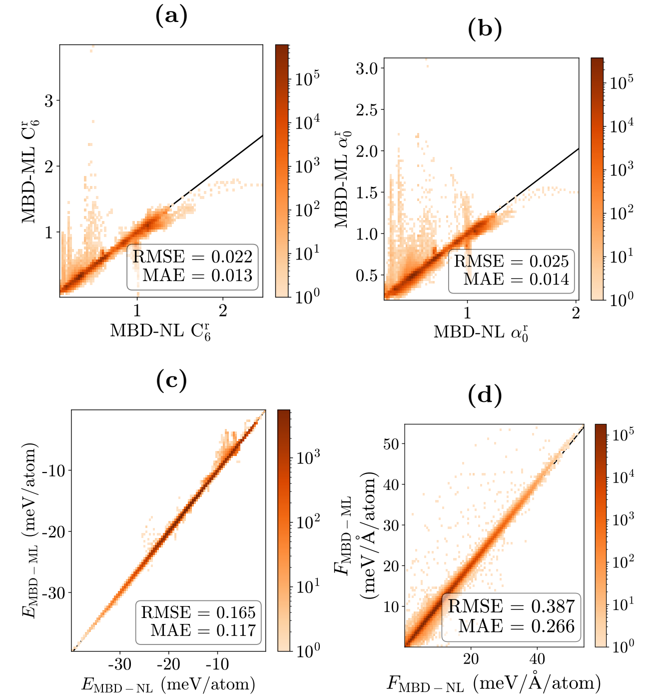


**Fig. S9.** Performance of PBE0+MBD-ML in predicting the $C_6$ and $\alpha_0$ ratios and the MBD contribution to the total energy and atomic forces of the DES370k test set excluding alkali and alkaline earth elements.

## S6. SENSITIVITY ANALYSIS OF MBD-ML

The present study demonstrates excellent accuracy of the MBD-ML model for nearly all target quantities, including relative energies, atomic forces, and stress tensor components. However, as noted in the main text and evident from Fig. 3, the total MBD energy predicted by MBD-ML can deviate from the *ab initio* reference by more than 100 meV for large systems containing over one hundred atoms. As shown in Table II, this discrepancy becomes negligible when computing physically measurable energy differences. Nevertheless, understanding the origin of this residual disagreement is instructive. In principle, exact prediction of the $\alpha_0^\mathrm{r}$ and $C_6^\mathrm{r}$ ratios would eliminate this error entirely.

To assess the sensitivity of the MBD energy to these inputs, we performed a sensitivity analysis for polymorph I of coronene. The MBD-ML predicted vdW energy for coronene I differs from the MBD-NL reference by $-61$ meV ($-0.85$ meV/atom), consistent with the MAE of the MBD-ML model on the OMC25 subset (Table I).

We first examined the impact of random, non-systematic noise in $\alpha_0^\mathrm{r}$ and $C_6^\mathrm{r}$ on the total MBD energy. Starting from the PBE0+MBD-NL reference ratios of coronene I, uniform zero-mean noise with a variable range between 0.0 and 0.04 was applied independently to both ratios. For each noise level combination, the MBD-NL energy was recomputed 100 times, and the resulting standard deviation was recorded (Fig. S10). The chosen range reflects the average prediction errors for these ratios on the OMC25 subset (Table I).

Perturbations within this range produce standard deviations in the total MBD energy of up to $\pm 38$ meV, substantially smaller than the $-61$ meV deviation observed for MBD-ML on coronene I. Based on Fig. S10 and the reported MAEs, the non-systematic contribution to the MBD energy error is estimated to be about 20 meV. The analysis also indicates that the MBD energy is slightly more sensitive to noise in $C_6^\mathrm{r}$ than in $\alpha_0^\mathrm{r}$, as evidenced by consistently larger deviations in the lower-left region of the heat map, although the difference amounts to only a few meV.

In addition to random errors, systematic bias must be considered. For the 200 molecular crystals in the OMC25 dataset, the mean residuals (serving as a proxy for bias) in Fig. 3 are $-0.02$ for $\alpha_0^\mathrm{r}$ and $-0.01$ for $C_6^\mathrm{r}$. To quantify the effect of such bias, the ratios for coronene I were independently shifted by values between $-0.04$ and 0.04 (Fig. S11). The resulting energy errors reveal a much stronger sensitivity to systematic shifts, with deviations reaching up to 700 meV and $-778$ meV. Again, shifts in $C_6^\mathrm{r}$ exert a larger influence on the vdW energy than comparable shifts in $\alpha_0^\mathrm{r}$.

Interestingly, energy errors are reduced when shifts in both ratios are of similar magnitude. Points along the diagonal of Fig. S11, where biases are balanced, exhibit smaller deviations. The observed biases of $-0.02$ and $-0.01$ correspond to an estimated MBD energy error of about $-44$ meV. Combining this systematic contribution with the estimated non-systematic uncertainty yields a total error estimate of $-44 \pm 20$ meV, consistent with the observed $-61$ meV deviation for coronene I.

Beyond rationalizing residual errors, this analysis provides guidance on the accuracy requirements for future models. Sensitivity estimates of this type can be used to determine the level of precision in predicted vdW ratios necessary to achieve a desired accuracy in the resulting MBD energies.

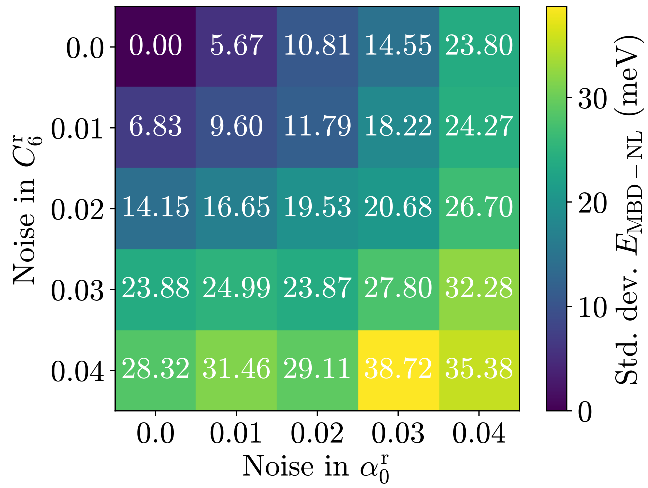


**Fig. S10.** Sensitivity of the MBD-NL energy to the introduction of uniform random noise to $\alpha_0^\mathrm{r}$ and $C_6^\mathrm{r}$.

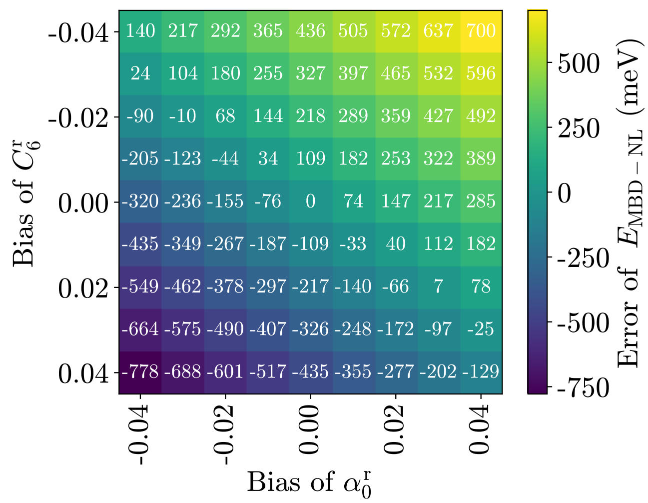


**Fig. S11.** Sensitivity of the MBD-NL energy to the introduction of a systematic bias to $\alpha_0^\mathrm{r}$ and $C_6^\mathrm{r}$.

## S7. DETAILS ON FORCE DEVIATION DISTRIBUTION FOR MBD-ML, D3 AND D4

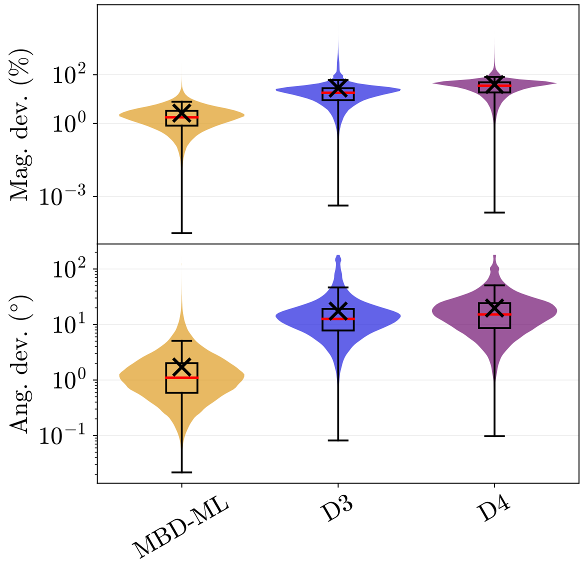


**Fig. S12.** Distributions of vdW force deviations from the MBD-ML, D3 and D4 methods compared to MBD-NL. The distributions are represented by violin plots. Black crosses correspond to the mean value, the red line to the median. The vertical edges of the boxes are the quartiles i.e., the 25-th and 75-th percentile of the deviation distribution. The whiskers correspond to the minimum and the 95% deviation.

Here we provide the quantitative statistical data on the distribution of the force magnitudes and angles of the MBD-ML, the D3 and the D4 method relative to the *ab initio* MBD-NL method. D3 and D4 vdW force contributions were computed using the `dftd3` (`https://github.com/dftd3/simple-dftd3`) and `dftd4` (`https://github.com/dftd4/dftd4`) packages. Both corrections were computed using the default PBE0 parameters, included Becke-Johnson (BJ) damping and an Axilrod-Teller-Muto three-body term and were compared to PBE0-based MBD-NL forces with a damping parameter $\beta = 0.83$. The same $\beta$ was used for the MBD-ML calculations. The distributions are illustrated via violin plots in Fig. S12, exact statistical values are given in Table S2.

**TABLE S2.** Mean, median, and 95$^{\text{th}}$ percentile deviations for relative magnitude and angular force predictions of MBD-ML, D3 and D4 compared to MBD-NL on the OMol25 test set.

| | MBD-ML | D3 | D4 |
|---|---|---|---|
| *Magnitude Deviation (%)* | | | |
| Mean | 2.6 | 28.2 | 39.5 |
| Median | 1.8 | 18.2 | 35.4 |
| 95$^{\text{th}}$ Percentile | 7.7 | 60.7 | 80.8 |
| *Angular Deviation (°)* | | | |
| Mean | 1.7 | 17.4 | 19.8 |
| Median | 1.1 | 12.6 | 15.2 |
| 95$^{\text{th}}$ Percentile | 5.1 | 46.6 | 50.8 |

**TABLE S3.** Molecular crystal polymorphs from which geometry optimizations were performed. Along with the roman numeral denoting the polymorph the CSD reference code and the type of interaction predominantly found in the system are provided: mixed, van-der-Waals (vdW) or hydrogen-bonded (hb).

| Molecule | Polymorphs | interaction type |
|---|---|---|
| aspirin | I (ACSALA07)<br>II (ACSALA17) | mixed |
| maleic acid | I (MALIAC12)<br>II (MALIAC13) | hb |
| 2-amino-5-nitropyrimidine | I (PUPBAD01)<br>II (PUPBAD02)<br>III (PUPBAD) | hb |
| maleic hydrazide | I (MALEHY10)<br>II (MALEHY01)<br>III (MALEHY12) | hb |
| pentacene | I (PENCEN)<br>II (PENCEN04) | vdW |
| coronene | I (CORONE04[a])<br>II (N/A[a]) | vdW |
| o-diiodobenzene | I (ZZZPRO06)<br>II (ZZZPRO07) | vdW |

[a] Structures directly taken from Ref. 16.

## S8. GEOMETRY OPTIMIZATION OF MOLECULAR CRYSTALS

For the structure relaxation of the molecular crystals the `Aims` calculator of the Atomic Simulation Environment (ASE) was employed. For geometry optimizations that involved the MBD-ML model, a new ASE calculator was implemented and combined with the Aims calculator as a `SumCalculator`, so that the final calculator object yielded total energies, atomic forces and stress tensors which were the sum of the pure FHI-aims output and the MBD-ML contributions to these properties. An example script to perform a PBE0+MBD-ML geometry relaxation can be found in the libmbd repository under `https://github.com/em819/libmbd/tree/master/examples/mbd-ml-ase/aspirin-structure-relaxation`. All geometry optimizations of molecular crystals were performed with the ASE implementation of the BFGS algorithm and involved two steps: Starting from the CSD reference structure, in the first step only the atomic positions were relaxed with a force threshold `fmax` of 0.005 eV/Å. The resulting structure was relaxed again, including both the atomic positions and the unit cell using the `FrechetCellFilter` without any constraints to an `fmax` of 0.005 eV/Å. In this second step, the initial guess for the Hessian in the ASE BFGS algorithm, `alpha` was chosen as 150.0 instead of the default 70.0 to counteract overshooting due to the notoriously shallow and rugged potential energy surface of molecular crystals, leading to more stable convergence. While the geometry optimizations were performed with the PBE DFA with MBD-NL, MBD-ML and without any vdW method, the final optimized structure was recomputed with the PBE0 functional instead. For PBE-based calculations that were combined with the MBD method (MBD-NL or MBD-ML), a damping parameter $\beta$ of 0.81 was used, while for the final PBE0 recalculations, a value of 0.83 was set.

**TABLE S4.** Structural comparison of polymorphs relaxed with PBE and PBE+MBD-ML relative to the PBE+MBD-NL-relaxed reference structures. RMSD and maximum atomic displacement (Max. Disp.) quantify the deviation of atomic positions, while $\Delta V$ gives the relative change in unit cell volume. Cases where the PBE geometry optimization did not converge are marked with "–". ANP refers to 2-amino-5-nitropyrimidine.

| | | RMSD (Å) | | Max. Disp. (Å) | | $\Delta V$ (%) | |
|---|---|---|---|---|---|---|---|
| Molecule | Polymorph | PBE | PBE+MBD-ML | PBE | PBE+MBD-ML | PBE | PBE+MBD-ML |
| *Mixed Interactions* | | | | | | | |
| Aspirin | I | 0.177 | 0.002 | 0.248 | 0.003 | 26.00 | 0.09 |
| | II | – | 0.007 | – | 0.018 | – | 0.22 |
| *Hydrogen Bonded* | | | | | | | |
| Maleic acid | I | 0.035 | 0.002 | 0.060 | 0.003 | 24.16 | 0.29 |
| | II | 0.040 | 0.001 | 0.077 | 0.002 | 27.19 | 0.19 |
| ANP | I | 0.222 | 0.004 | 0.385 | 0.006 | 28.32 | 0.22 |
| | II | 0.401 | 0.002 | 1.046 | 0.004 | 37.47 | -0.18 |
| | III | 0.062 | 0.001 | 0.149 | 0.001 | 13.22 | 0.05 |
| Maleic hydrazide | I | 0.055 | 0.001 | 0.078 | 0.002 | 28.13 | 0.05 |
| | II | 0.170 | 0.003 | 0.196 | 0.007 | 30.76 | -0.11 |
| | III | 0.042 | 0.001 | 0.068 | 0.002 | 23.23 | 0.00 |
| *van der Waals* | | | | | | | |
| Pentacene | I | 0.176 | 0.004 | 0.293 | 0.008 | 36.18 | -0.46 |
| | II | – | 0.007 | – | 0.014 | – | -0.55 |
| Coronene | I | – | 0.004 | – | 0.007 | – | -0.26 |
| | II | 0.277 | 0.001 | 0.499 | 0.002 | 37.42 | -0.18 |
| *o*-Diiodobenzene | I | 0.119 | 0.005 | 0.179 | 0.008 | 29.86 | -0.50 |
| | II | 0.085 | 0.038 | 0.119 | 0.056 | 26.65 | 0.12 |

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